CMSC 420: Bloom Filters

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1 Problems **2**

1 Problems

1. Suppose we define a Bloom filter using $S = \mathcal{Z}_{12}$, with B indexed by $\mathcal{Z}_{11}$, and with two hash functions:

$$h_1(x) = 2x \mod 11$$
$$h_2(x) = 3x \mod 11$$

   (a) Suppose you insert the keys $x = 1, 6, 9$. Show the state of the Bloom filter after each insertion.

   (b) Identify all false positives.

2. Suppose we define a very small Bloom filter with $m = 3$ and $k = 2$.

   (a) We insert $n = 1$ keys. Write down all the possible resulting bit arrays and the probability of each.

   (b) We insert $n = 2$ keys. What is the probability that the resulting bit array is `100`?

   (c) We insert $n = 2$ keys. What is the probability that the resulting bit array is `110`?

3. Suppose we define a Bloom filter with $m \geq 2$ and $k = 2$. We insert $n = 2$ keys.

   (a) What is the probability (in terms of m) that the resulting bit array has `1`s in exactly one index and `0`s elsewhere?

   (b) What is the probability (in terms of m) that the resulting bit array has `1`s in exactly two indices and `0`s elsewhere?

4. Using the inaccurate (but not terrible) bloom filter false positive expression:

$$p = \left(1 - \left(1 - \frac{1}{m}\right)^{kn}\right)^{k}$$

   (a) Suppose we wish to store $n = 70$ items with $k = 3$ hash functions and a bit array of size $m = 50$. What would the corresponding false positive probability be? Give an exact answer (unsimplified) and an approximation to four decimal digits.

   (b) Suppose we wish to store $n = 100$ items and we want a false positive rate no higher than 2%. Find a restriction on m (as a function of k) which will guarantee this.

5. Suppose the state of a particular count array for a Counting Bloom Filter after n insertions with k hash functions is:

```
B = [3,1,2,1,0,2,2,0,0]
```

Suppose you don't know k but you know that n is a single-digit number. Moreover suppose that for any given key there are at least two hash functions which produce different outputs. Prove that there are at least six distinct hash functions.

6. Suppose the state of a particular count array for a Counting Bloom Filter after n insertions with k hash functions is:

```
B = [2,1,2,2,1,5,1,0,0]
```

Suppose you know both k and n are single-digit numbers and you know that for any given key no more than two of the hash functions will give identical results.

What are n and k? Justify.

7. Suppose we define a Bloom filter with $m = 4$ and $k = 3$. We insert $n = 2$ keys.

   What is the probability that the resulting bit array has `1`s in exactly two indices and `0`s elsewhere?

8. Suppose we define a very small Bloom filter with $m = 4$ and $k = 2$.

   (a) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1000`? Show your work.

   (b) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1111`? Show your work.

   (c) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1010`? Show your work.

9. Suppose we define a Bloom filter with $m = 3$ and $k = 2$. We insert $n = 2$ keys.

   What is the probability that the resulting bit array is `111`?

10. Suppose a Bloom Filter has table size $m = 1000$ and we use $k = 50$ hash functions. What is the maximum number of keys we can store in it to ensure that the probability of a false positive is at most 0.9? Show your calculation and only approximate at the final step.

    Use the incorrect (but simple and approximate) formula for this.

11. Consider the incorrect (but simple and approximate) formula for a false positive. For fixed $m = 100$ and $n = 50$ plot this as a function of k for $0 \leq k \leq 20$. Comment on why its shape makes intuitive sense.

12. Suppose we define a very small Bloom filter with unknown $m = 3$ and $k = 2$. We insert $n = 2$ keys. What is the probability that the resulting bit array is `011`? Show your work.